Mino: Money In, Noise Out

A Centralized, Gamified Platform for Building Strong Financial Habits in Young Adults

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Abstract

Mino is a mobile personal finance application designed to help users build better financial habits through guided budgeting, an onboarding habits quiz, and short financial literacy lessons. Building on the interactive Figma prototype developed in earlier phases, this report presents the Phase IV usability evaluation, using moderated, think-aloud usability testing across three core task flows: completing the onboarding quiz, creating a category-based budget, and completing an investment lesson. Participants' navigation patterns, task completion outcomes, and verbal feedback were recorded using a structured observer data sheet, followed by a standardized debrief interview and post-task questionnaire to capture subjective ease-of-use ratings and open-ended feedback. Results indicate strong overall learnability, with participants navigating more efficiently across later tasks after becoming familiar with the sidebar-based structure. However, testing surfaced several recurring points of friction, including the absence of a back/edit function within the quiz, ambiguity around budget category time frames, and limited discoverability of lesson topics without scrolling. These findings coupled with our past findings from the previous phases are used to derive concrete design recommendations aimed at improving navigational clarity, input affordances, and information visibility within the prototype, informing Mino's next iteration.

1 Introduction

Financial literacy remains a persistent challenge for young adults, many of whom enter adulthood without practical experience managing budgets, savings, or long-term financial goals. This gap can lead to poor spending habits and financial stress later in life. Currently, existing personal finance apps address this gap only partially. Many offer budgeting tools to track spending, but few pair that tracking with structured education or personalized insight into a users' financial behavior. The objective of our project is to apply Human-Computer Interaction principles to develop *Mino*, a mobile application designed to improve the financial literacy of young adults. Mino includes three core features, a personalized financial literacy onboarding quiz, a budgeting tool, and bite-sized educational lessons. The onboarding quiz evaluates a user's financial habits to provide a specified personality type (e.g., "The Architect-Saver" or "The Spark-Impulsive Spender"), and to help the user better understand their spending behaviors. The budgeting feature enables users to monitor their finances and work toward their financial goals, while the bite-sized learning section offers accessible lessons on a variety of financial topics. Throughout the previous project phases, we established the application's concept, gathered user requirements through surveys and focus groups, and designed a high-fidelity prototype. Building on the evaluation plan developed in Phase III, we have evaluated the prototype with our target audience as participants in our usability testing phase, and have used the resulting feedback to refine and finalize the application prototype.

2 Literature Review

Young adults are increasingly expected to make important financial decisions despite often lacking the foundational knowledge needed to make those decisions.

Lusardi, Mitchell, and Curto (2010) examined financial literacy among young adults using data from the National Longitudinal Survey of Youth and found that only 27% of participants demonstrated basic knowledge of interest rates, inflation, and risk diversification. This knowledge gap is especially important because young adults may already be making important financial decisions involving student loans, credit card debt, saving, and other long-term financial commitments before fully understanding how those financial tools work.

Adesina, Carnaghan, and Smith similarly examined financial literacy among young adults in Canada and the United States and found that participation in financial education does not necessarily translate into stronger financial literacy. Their review of Canadian Financial Capability Survey data showed that although 56% of Canadians aged 18–34 reported engaging in financial learning, many of them did not demonstrate major improvements in financial literacy. They also identified a similar age-based literacy gap in U.S. survey data, where young adults answered fewer than 46% of financial literacy questions correctly compared with 54% correct answers among adults aged 60 and older. The authors further argue that many financial education programs lack a strong theoretical foundation, making their long-term effectiveness and behavioural retention uncertain.

Together, these studies establish the central problem Mino aims to address. Not only do young adults need greater access to financial information, but financial education that can be understood, retained, and applied to real financial decisions. The consequences of limited financial literacy extend beyond knowledge and can influence students' stress, confidence, and long-term well-being.

Danahy, Loibl, Montalto, and Lillard provide large-scale evidence of this relationship through their study of more than 25,000 college students. They found that students with greater student loan debt and lower emergency savings reported higher levels of financial stress. Importantly, students with stronger financial socialization and financial self-efficacy experienced lower levels of stress, indicating that knowledge and confidence in financial decision-making may improve students' ability to deal with financial responsibilities. Their research also highlights longer-term consequences, as students carrying loan debt may delay major life decisions such as marriage, having children, and purchasing a home.

Archuleta, Dale, and Spann (2013) similarly examined the relationship between debt, financial satisfaction, and financial anxiety and found that financial satisfaction accounted for approximately 30% of the variance in reported financial anxiety. Together, these findings show that financial management is closely connected to someone's emotional well-being. This supports Mino's goal of helping users actively manage their finances through budgeting guidance and manageable financial tasks rather than only presenting financial information. Research into how financial education is delivered supports Mino's use of short, and engaging lessons.

Zhao (2025) surveyed 100 University of British Columbia students and found that traditional financial education was viewed as too theoretical and difficult to apply, with only 30% of students who had taken formal finance courses reporting that the courses improved their financial decision-making and more than half saying they forgot most of the material within a year. In contrast, more than 90% of students said they would rather view a short financial explanation when it was relevant to their situation, and all participants said just-in-time education is more useful than a traditional course format.

These findings align with Mbonigaba Celestin and N. Vanitha's research on gamification in personal finance applications, which found that gamified features increased user engagement by 45% and were associated with a 30% increase in savings among younger users. However, their study also showed that greater engagement and confidence did not necessarily lead to greater financial literacy, as user confidence increased by 25% while literacy scores remained relatively unchanged, and 40% of novice users moved toward higher-risk investments without sufficient knowledge. Together, these studies support Mino's decision to combine bite-sized financial lessons with engaging and interactive elements, while also showing that these features should be designed to reinforce actual learning rather than simply increase participation or confidence. Research also supports Mino's decision to combine multiple financial tools within a single application rather than separating education, budgeting, and financial management into different experiences.

French et al. (2020) examined an experiment in which participants were given access to four financial applications, including a loan-interest comparison tool, an expenditure comparison tool, a cash calendar, and a debt management application. Participants who received these tools demonstrated greater financial resilience and were more likely to rely on their own resources when faced with an unexpected expense. The study shows that practical digital tools can help users translate financial knowledge into everyday financial management. This supports Mino's all-in-one approach, where users can access budgeting tools, financial education, and personalized guidance within the same application. In particular, Mino's category-based budgeting feature gives users a practical way to apply what they learn while monitoring their spending and working toward financial goals, rather than requiring them to move between separate financial resources. Personalization may strengthen financial education by recognizing that users approach money differently and may require different forms of guidance.

Luksander, Németh, and Zsótér (2017) developed a 36-item financial personality assessment and identified seven distinct financial personality types within their sample. They also linked these personality types to different patterns of personal indebtedness using a decision-tree model. Their findings suggest that financial behaviour can cluster into recognizable behavioural profiles rather than existing along a single scale. This provides a foundation for Mino's onboarding quiz, which similarly categorizes users into financial personality archetypes to encourage reflection on their habits.

Tovanich et al. (2021) approached personalization from a behavioural perspective by analyzing transaction records from 1,306 bank customers and examining whether psychological characteristics could be inferred from spending patterns. Their findings suggest that a user's financial profile may be informed not only by an initial questionnaire but also by ongoing financial behaviour. Together, these studies support Mino's decision to personalize guidance based on financial archetypes while also allowing those archetypes to evolve over time as a user's spending patterns and behaviours change.

Overall, the literature identifies both the problem Mino is attempting to solve and the principles that should guide its solution. This research shows that young adults often lack fundamental financial knowledge while simultaneously facing increasingly complex financial responsibilities that can contribute to stress, anxiety, and long-term consequences. At the same time, the literature shows that simply making financial information available is insufficient. Effective financial education should be practical, understandable, timely, engaging, and connected to real financial behaviour. These findings provide research-based support for Mino's core features: the onboarding personality quiz is supported by research on differences in financial behaviour, the budgeting tool provides an opportunity to actively apply financial knowledge and access other tools in one platform, and bite-sized lessons that help users apply and retain financial topics of their choice.

3 Problem Statement

Financial literacy remains a widespread gap among young adults, as only 27% of young people in one study could demonstrate basic understanding of concepts like interest rates, inflation, and risk diversification (Lusardi et al., 2009). This gap carries real consequences, since early financial decisions, like taking on student loans or credit card debt, shape a person's long-term financial stability, often causing overwhelming stress before they fully understand the financial tools they are using. The problem is not a lack of financial education programs, but rather many existing programs fail to produce retained behavioural change because they are not grounded in a clear theoretical framework (Adesina et al., 2024). Post-secondary students are a particularly important group to focus on, since they are reaching a life stage where financial responsibilities intensify simultaneously (student loans, rent, utilities, groceries) while their financial literacy has not necessarily caught up. This is reflected in survey data showing that college students carrying loan debt report higher financial stress, particularly when they lack emergency savings, and that financial stress in this population extends beyond immediate money management into decisions about relationships, housing, and family planning (Danahy et al., 2024). Financial strain during this period is also tied to broader emotional wellbeing: financial dissatisfaction alone accounts for roughly 30% of the variance in reported financial anxiety among students (Archuleta et al., 2013). Mino was designed to address this gap directly. Mino is a mobile application that helps post-secondary students recognize their own financial personality, build sustainable budgeting habits, and learn financial concepts in a format suited to their day-to-day lives.

3.1 Research Question

This raises our guiding research question for Phase IV: to what extent does Mino's onboarding quiz, budgeting tool, and bite-sized lesson format actually support post-secondary students in building financial understanding and confidence, and where does the current prototype fall short in usability terms that would need to be addressed before further development?

4 Gathering of User Requirements

To figure out what Mino actually needed to do, our team relied on two techniques that complemented each other: a survey to get breadth across a large pool of post-secondary students, and an online focus group to test specific feature ideas in more depth. The survey told us what students in general think and struggle with. The focus group told us whether our proposed solutions actually made sense to them.

4.0.1 Survey. We collected 51 responses from post-secondary students aged 17 to 24, recruited mostly through the r/Survey subreddit along with Instagram and personal connections, drawing from UTSC, TMU, Ontario Tech, Guelph, and a few other schools. The 16-question survey used a consistent 1 to 10 scale throughout, rather than a 1 to 5 scale, since a wider scale gives clearer separation between answers. Every question was also tested by five observers to make sure it met Fink's (2003) definition of measurable, meaning two or more people would interpret it the same way. Two key findings emerged from this data. First, a budgeting and cost-saving gap: 88% of respondents rely on someone else for housing, but 62% still pay over a quarter of their own expenses, and most rated their effort to save money as moderate to high (24% picked 7 out of 10). This implies money management is already a real part of participants' lives without much support behind it. Second, an investing confidence gap: over 60% said they are at least moderately interested in investing, yet 75% of that group rated investing as a 5 or higher in difficulty, and only half of the interested group actually invests. On top of that, 75% of respondents named investing as a topic they would want to learn more about in the open-ended wrap-up question. That gap became our main justification for building an educational, low-risk way to learn about investing.

4.0.2 Focus Group. After the survey, we ran an online focus group with 8 participants for about 35 minutes, walking them through a low-fidelity wireframe of the financial personality, budgeting, and bite-sized learning features, then opening the floor up for discussion. The biggest takeaway was that participants preferred an in-app investment simulator over either being redirected to an outside investing app or being allowed to invest real money right away. Several people felt that jumping straight into real investing after one lesson would be overwhelming, and that a simulator gave them room to practice without real risk. Participants also responded well to gamification (XP, streaks), but expressed some resistance to the idea of being locked into a single, fixed "financial personality" label, and asked for more clarity on when and how the AI inside the app was actually influencing their budget or goals.

4.0.3 Personas and Task Analysis. To keep our design grounded in real variation rather than one imagined user, we built three personas: Maya (a financial procrastinator who is supported by her parents), Daniel (an impulsive, gambling-leaning beginner investor with irregular co-op income), and Andrea (a frugal economizer paying her own rent). Using these personas alongside our survey and focus group data, we developed hierarchical task analyses diagrams covering onboarding, budgeting, and the learning roadmap, and narrowed things down to five key user tasks: completing an onboarding quiz to get a financial archetype, setting up a budget with optional AI help, adding financial goals, paper-trading through an investment simulator, and completing knowledge-check quizzes after lessons.

4.1 Types of Users

Our target users are post-secondary students in Ontario, ages 18–25, who generally do not feel confident managing their own money, though the group spans both students financially supported by parents and those covering their own expenses. We grouped users along four dimensions: level of app use (novice, expert, frequent, casual, and users needing extra accessibility support), financial experience (novice, intermediate, advanced), income type (no income, part-time, full-time, seasonal, government-funded), and living situation (with parents/guardians, living alone, with roommates, or transitional housing). This allows us to design for both a first-time user just opening a bank account and a more experienced user already investing, without assuming either is the "default."

4.2 Types of Requirements

Together, these findings pointed to three types of requirements: functional, non-functional, and user experience. Functional Requirements: Mino should keep the investment simulator locked until the user completes the required lessons, then unlock it automatically. Mino should let users create and edit a budget in real time, including adding custom spending categories beyond the default set. Mino should track lesson progress cumulatively (a gradient/heatmap), rather than a linear streak that resets after a single missed day. Non-Functional Requirements: Mino should make the AI's reasoning visible to the user wherever it acts on their behalf, such as budget suggestions or goal recommendations, so users can trust what it is doing. User Experience Requirements: Mino should present a user's financial identity as a percentage distribution across archetypes, rather than a single fixed label, so users do not feel inaccurately boxed into one category.

5 Prototype

Based on these requirements, we built a high-fidelity, interactive prototype in Figma for a mobile application. Given our time constraints, we focused on making three core features fully interactive: the financial archetype quiz, budget management, and bite-sized learning. Other features, like recurring payment management, the full XP goal system, and the investment simulator, are only partially built out for now. All three main tasks are reachable from a bottom navigation bar (home, budget, lessons, profile) once a user finishes onboarding.

5.1 Task 1: Onboarding Quiz

New users go through an eight-question, scenario-based quiz (for example, "You just got paid, what's your first move?") that samples spending, saving, risk tolerance, and planning behaviour. We went with scenario-based questions instead of abstract self-ratings because people tend to answer more honestly when picturing a real situation rather than rating themselves in the abstract, which cuts down on social-desirability bias. Instead of assigning one label, the result screen shows a percentage breakdown across all the archetypes (Figure 1), directly answering the focus group's concern about being boxed into one category. Users can retake the quiz or skip it and come back later, which keeps them in control of the process, in line with Shneiderman et al.'s (2017) principle of user control and freedom.

Figure 1: Onboarding quiz result screen, showing a percentage breakdown across archetypes rather than a single fixed label.

5.2 Task 2: Creating a Budget

From the budgeting tab, users set up a budget for whatever period fits them (weekly, monthly, annual, or one-time), entering their income and default categories like grocery, transportation, bills, leisure, and savings, or adding a custom category like "Vacation." Once it is activated, the dashboard shows a pie chart and per-category progress bars comparing what has been spent to what was planned (Figure 2), so the system's status is always visible. This turns budgeting into something concrete and ongoing rather than an abstract idea, and it speaks directly to the budgeting gap our survey turned up.

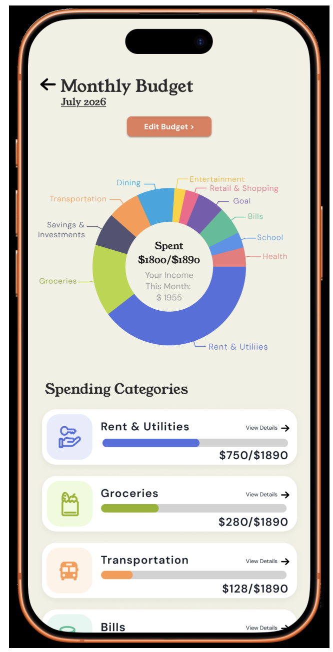

Figure 2: Active budget dashboard, showing spend-versus-plan by category.

5.3 Task 3: Financial Lessons

From the lessons tab, users pick a category (Investing, for example), then choose a topic (like "Stocks vs. bonds, ETFs"), and go through a lesson broken into three phases: a simple definition of the topic, a real-life application example of this topic, and a one-question knowledge check, before landing back on a roadmap showing what unlocks next. Keeping each lesson to small, single-purpose screens follows Shneiderman et al.'s (2017) idea of reducing short-term memory load. Finishing lessons gradually unlocks parts of the paper-trading simulator (Figure 3), so users only get access to riskier practice once they have actually engaged with the lessons behind it. This was the design decision the focus group responded to most positively.

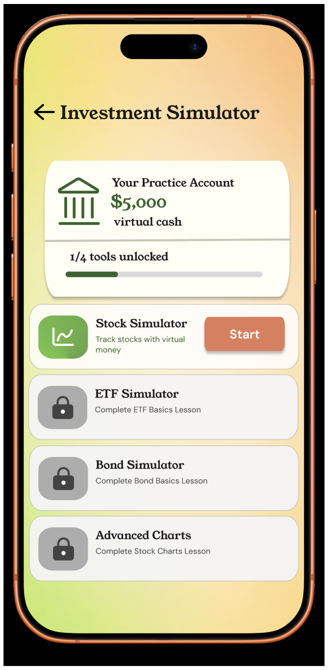

Figure 3: Investment simulator, showing the practice account and tools that unlock progressively as lessons are completed.

5.4 Overall Design Approach

Across all three tasks, we used colourful, consistent visual components and an avatar-driven tone on purpose to keep the experience approachable for students who told us they do not feel confident managing their finances. At the same time, the underlying interactions, like editable budgets, retakeable quizzes, and regenerable quiz questions, were built to keep users in control at every step (Shneiderman et al., 2017).

6 Usability Study

A moderated usability study was conducted to determine whether Mino's target users would be able to complete and understand the three main tasks represented in the high-fidelity prototype. The study used a mixed-method approach that combined think-aloud testing, structured observation, pre-test and post-test questionnaires, and a standardized debrief interview. This method was chosen because it offered the chance to obtain both qualitative and quantitative data from the test. Quantitative data such as task completion status, task completion time, error rate and assistance were used to measure participants' performance of the tasks, while qualitative data including think-aloud comments, observer notes, open-ended questionnaire responses, and debrief feedback, were used to determine why participants had difficulties. Using multiple data-gathering techniques also supported triangulation, providing better connective evidence for observed participant behaviour.

6.1 Participants

Participants were chosen from the target population of Mino: post-secondary students with different financial literacy backgrounds and experiences. This population was selected because Mino is designed to support students who are beginning to manage increasingly complex financial responsibilities. Before participating, each participant received an informed-consent document, which provided information on the aim of the study, the procedure, the voluntary nature of the task, and the participant's right to withdraw at any time. Additionally, participants were also given an opportunity to ask questions before beginning.

Next, participants were asked to complete a pretest questionnaire regarding their experience with technology, budgeting and financial literacy. The questionnaire provided background information that could be used to help further interpret participant behaviour during the usability session.

6.2 Testing Environment

The test was carried out using a combination of in-person and remote moderation testing. When the test was remote, participants shared their screen with the moderator to observe their interaction with the prototype. All participants interacted with the exact high-fidelity prototype created on Figma and completed the same three pre-established tasks in the same order. The moderator gave all the participants the same standard task instructions and observed their interactions with the prototype. No step-by-step instructions were given to participants as the goal of the study was to see whether the prototype communicated what needed to be done to perform each task. The study can be defined as usability testing conducted in a controlled temporary environment. While there were variations in physical location and device of the participants, other variables were controlled such as the prototype used, the task order, task instructions, and data collection method.

6.3 Testing Procedure

Before every session began, the moderator would introduce the study and confirm that the subject had filled out the consent form and the pre-test. Then the subject would be told about the objective of the first task. Participants were encouraged to use the think-aloud technique during the entire test session. More specifically, they were asked to describe what they were looking for, what they expected to happen after selecting an option, and anything they found unclear or confusing. During each task, the moderator or note-taker recorded: task-completion status task-completion time errors and misclicks hesitation requests for assistance relevant verbal comments Once participants had gone through all three scenarios, they were asked to complete a post-test survey that included both rating scale and open-ended questions. Thereafter, participants underwent a debriefing process that enabled the researcher to get their overall impression, identify which scenario confused them the most, discover any missing elements or pieces of information, and their highest priority recommendation.

6.4 Evaluated Tasks

6.4.1 Task 1: Onboarding Quiz. The first task required participants to complete the onboarding quiz and view their assigned financial personality. The task evaluated whether participants could understand the purpose of the quiz, select responses, navigate between questions, and complete the flow without unnecessary assistance.

The moderator noted if the participant identified the correct controls, understood the questions and its answers, and knew when he or she had successfully completed the quiz. Time taken to complete, task completion status, errors, hesitation, and help sought were all recorded as well.

6.4.2 Task 2: Creating a Budget. The second task required participants to create a monthly budget by entering limits for different spending categories and then interpreting the resulting budget dashboard. It was aimed at assessing whether participants could identify budgeting features, comprehend the budget creation process, set category values, and interpret the information displayed on the completed dashboard. Specific focus was put on how participants could understand the total budget value, money left to spend, spending categories and timeframe represented by the interface.

6.4.3 Task 3: Financial Lesson. The third task consisted of finding and completing one investment lesson, answering its knowledge check and examining the correlation between the completion of the lesson and the investment simulation. This task assessed the ability of the participants to navigate the lesson categories and roadmap, start the appropriate lesson, work through the lesson content, complete the knowledge check, and learn about lessons, XP, progress and simulator access.

6.5 Usability Measures

The study evaluated five usability goals: learnability efficiency errors and recovery memorability satisfaction Learnability was investigated through task success, the level of guidance needed, and users' understanding of the interface in their first attempt. Efficiency was measured primarily through completion time and perceived unnecessary steps. Errors and recovery were assessed by recording mistakes, misclicks, requests for assistance, and questionnaire ratings related to recovering from errors. Memorability was assessed through participants' perceptions of whether they would be able to complete the lesson flow more easily during a future attempt. Satisfaction was evaluated using rating-scale responses, open-ended comments, and the standardized debrief.

7 Results

7.0.1 Participants. Twelve participants (N = 12) completed the post-task questionnaire following the usability session. Of these, ten sessions (n = 10) included complete observer-recorded task performance data (task completion status, duration, and behavioural checklist items); the remaining two sessions had partial or unspecified observer records and are excluded from the observer-based statistics below. Participant identifiers (P1–P12) are used throughout this section in place of participant names to preserve anonymity. Percentages based on observer data use n = 10 as the denominator; percentages based on questionnaire data use N = 12. Note on prototype fidelity: because the Figma prototype was configured to support a faster walkthrough, each quiz screen exposed only one interactive option rather than the full set of intended selectable answers. This is a limitation of the prototype's interactivity rather than of the underlying design and is a likely contributor to some of the input-related friction reported below.

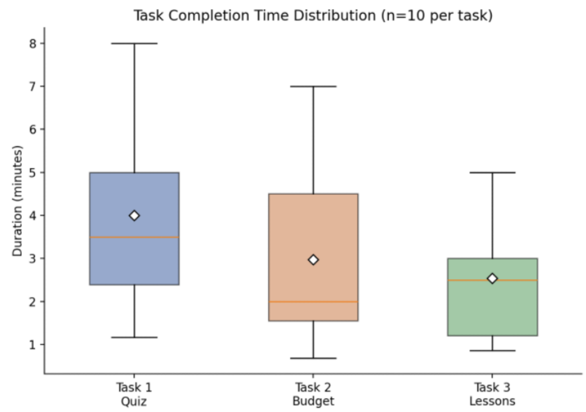

Figure 4: Distribution of task completion times for Task 1 (Quiz), Task 2 (Budget), and Task 3 (Lessons), n = 10 observed sessions per task.

7.1 Task 1: Onboarding Quiz

Of the ten sessions with a recorded observer outcome, 7/10 participants (70%) achieved Direct Success on the onboarding quiz task, 2/10 (20%) achieved Indirect Success (requiring minor guidance or recovering from a misclick), and 1/10 (10%) was not specified by the observer. No failures or task abandonment were recorded. Observers also noted that 10/10 (100%) of participants located the 'quiz flow' entry point in the sidebar without difficulty, and 9/10 (90%) were judged to understand the quiz's purpose (i.e., that it assessed financial habits to personalize future recommendations). This suggests that navigation to the task and comprehension of its intent were not meaningful barriers. By contrast, 5/10 (50%) of participants were observed struggling with quiz inputs or controls at some point, most often around selecting an intended multiple-choice option, indicating that interaction-level friction, rather than navigation or comprehension, was the primary source of difficulty in this task. Task 1 completion time averaged 4.00 minutes (SD = 2.24, median = 3.5 minutes, n = 10), ranging from approximately 1.2 to 8 minutes. This spread is wide relative to the mean, suggesting substantial variation between participants in pacing and think-aloud engagement rather than a single consistent difficulty level.

7.1.1 Self-Reported Ratings. On the post-task questionnaire (7-point scales unless noted), participants rated the quiz moderately positively overall. Perceived sufficiency of on-screen information averaged M = 5.33 (SD = 2.06), and quiz length appropriateness averaged M = 5.75 (SD = 1.86), together suggesting that most participants found the amount of information and number of questions acceptable rather than excessive. Perceived accuracy of the quiz in evaluating one's financial personality was rated lower, at M = 4.92 (SD = 1.93), the weakest of the four rated dimensions; this indicates that while participants could complete the quiz comfortably, a portion remained unconvinced that its output meaningfully captured their financial personality. Ease of recovering from an unintended selection averaged M = 5.75 (SD = 1.76), suggesting that when missteps did occur, most participants did not find them costly to correct. A separate item asked participants to rate how easy it was to understand the quiz's buttons and controls, using a reverse-scored scale (1 = very easy). Responses were tightly clustered at M = 1.42 (SD = 0.51), indicating that participants generally found the controls themselves easy to understand. Taken alongside the 50% input-struggle rate noted by observers previously, this may indicate that the friction participants encountered was momentary or specific to individual screens (e.g., a single unresponsive option, consistent with the prototype constraint noted previously) rather than a broader problem with the controls' design; however, this study did not directly test that explanation.

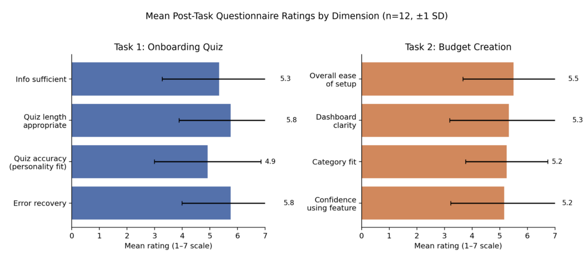

Figure 2. Mean self-reported ratings (±1 SD) for Task 1 and Task 2 dimensions, N = 12. Scale: 1–7, higher = more favourable, except "Buttons Ease" (not shown; reverse-scored, M = 1.42).

Figure 5: Mean self-reported ratings (±1 SD) for Task 1 and Task 2 dimensions, N = 12. Scale: 1–7, higher = more favourable, except "Buttons Ease" (not shown; reverse-scored, M = 1.42).

7.1.2 *Recurring Issues.*

- Unresponsive quiz options: one participant reported being "unable to choose the button/option I wanted to... limited to one option that did not represent what I wanted to pick." This is consistent with the single-interactive-option prototype constraint described previously, and may not reflect an issue with the intended design.
- No back/edit navigation: participants across sessions expected to revisit or change a previous quiz answer and found no mechanism to do so.
- Near-duplicate answer choices: one participant noted that a question about financial-future outlook "had two answers that felt very similar and expressed the same thing," creating ambiguity in the intended response.
- Minor readability issues: small font size was flagged once as a mild source of friction.

7.2 Task 2: Creating a Budget

Among the ten observed sessions, 5/10 participants (50%) achieved Direct Success on the budgeting task, 4/10 (40%) achieved Indirect Success, and 1/10 (10%) was not specified. As with Task 1, no failures or abandonment were recorded, but the Direct Success rate was lower than Task 1's 70%, indicating that this task more often required some form of guidance or recovery. All 10/10 (100%) observed participants located the 'Budget Flow' entry point without difficulty, and 8/10 (80%) were able to set category spending limits smoothly, suggesting that the core input mechanics of budget creation were generally usable. Understanding of the resulting spending dashboard was weaker: only 5/10 (50%) of observed participants were judged to fully understand the dashboard summary and "spare amount" display, while the remainder showed partial understanding, required a clarifying question, or did not understand it at all. This gap between smooth input (80%) and full dashboard comprehension (50%) suggests that the primary usability bottleneck in Task 2 was interpreting the output of the budgeting flow rather than performing the budgeting actions themselves. Task 2 completion time averaged 2.98 minutes (SD = 2.23, median = 2.0 minutes, n = 10), which was shorter than Task 1's mean of 4.00 minutes (Figure 1). This difference may reflect increased familiarity with the prototype's navigation after completing the first task, although this study was not designed to isolate a learning effect from other factors (e.g., Task 2 may simply have required fewer discrete steps), so this interpretation should be treated as tentative.

7.2.1 *Self-Reported Ratings.*
Self-reported ease of setting up the budget from start to finish averaged M = 5.50 (SD = 1.83), and confidence in using the budgeting feature going forward averaged M = 5.17 (SD = 1.95), together suggesting a moderately positive but not unreserved experience. Clarity of the spending dashboard averaged M = 5.33 (SD = 2.15) with the widest spread of any Task 2 item, mirroring the mixed dashboard-comprehension outcomes observed previously and reinforcing that dashboard interpretation was a more variable experience across participants than the budget-setup steps themselves. Fit of the default budget categories with participants' own money-categorization habits averaged M = 5.25 (SD = 1.48). Notably, 0/12 participants (0%) reported making an outright budgeting mistake when asked directly, even though observer notes and the lower dashboard-comprehension rate point to real points of confusion; this discrepancy suggests that some usability issues (e.g., unclear totals) were not registered by participants as "mistakes" on their part, rather than that no such issues occurred.

7.2.2 *Recurring Issues.*

- Category total mismatch: at least two participants independently flagged that the sum of individual category spending did not appear to match the total spend figure shown at the top of the dashboard.
- Ambiguous time frame: budget inputs did not specify whether limits were weekly or monthly, leading to uncertainty when entering amounts.
- Save-state confusion: one participant's session transitioned to a blank "Edit Budget" screen with a "Clear Budget" button after saving, leaving it unclear whether data had been saved or reset; the same participant subsequently and unintentionally cleared their entries, and a second, independent participant reported a similar accidental budget-clearing incident.
- Zero-value spend fields: "Spend Budget" columns displayed $0 for most categories despite real spending amounts being entered, undermining trust in the dashboard's accuracy.

- Category granularity: one participant suggested clearer visual separation (e.g., colour-coding) between categories, and another requested a direct comparison between actual and expected spending per category.

7.3 Task 3: Financial Lessons

7.3.1 Task Completion. Of the ten sessions with a recorded observer outcome, 9/10 participants (90%) achieved Direct Success on the financial lessons task, and 1/10 (10%) was not specified by the observer. No Indirect Successes, failures, or abandonment were recorded, making this the highest and most consistent completion rate of the three tasks. Observers further noted that all 10/10 (100%) participants located the 'Lessons Flow' entry point in the sidebar, one using non-standard but clearly positive phrasing ("This is really good") rather than a plain "Yes", and all 10/10 (100%) completed the full lesson screen flow once started. Starting the investment lesson itself was also largely frictionless, with 9/10 (90%) proceeding directly and 1/10 (10%) experiencing a slight delay locating it. Taken together, this pattern suggests that once participants reached the lessons area, both navigation and task completion were highly reliable; the main source of variation was locating the correct entry point within the section rather than completing the lesson itself. Task 3 completion time averaged 2.54 minutes (SD = 1.55, median = 2.5 minutes, n = 10), ranging from approximately 0.9 to 5 minutes, the fastest average of the three tasks (Figure 1). Given that Task 3 was also completed last in the session sequence, this may again reflect accumulated familiarity with the prototype's navigation pattern, consistent with (and tentatively reinforcing) the same caveated interpretation offered for Task 2.

7.3.2 Self-Reported Ratings. On the post-task questionnaire, participants rated the lesson's experience positively overall, and generally more favourably than either the quiz or the budgeting task. Ease of lesson navigation averaged M = 5.33 (SD = 1.87), and appropriateness of the information presented per lesson averaged M = 5.75 (SD = 1.14), suggesting the lesson content itself was perceived as digestible. Confidence navigating the lessons more easily next time was the highest-rated item across all three tasks at M = 6.58 (SD = 0.51), with very low variance, indicating that whatever initial friction participants experienced, they were confident it would not recur, which is consistent with a one-time discoverability issue rather than a persistent design flaw. Ease of error recovery averaged M = 6.42 (SD = 0.67) and overall satisfaction with the bite-sized lesson format averaged M = 6.25 (SD = 0.97), both comparatively high. The lowest-rated Task 3 item was whether earning XP points felt rewarding, at M = 5.17 (SD = 1.59), suggesting the gamification element was less consistently compelling than the core lesson content itself.

7.3.3 Recurring Issues.

- Unclear purpose of XP/reward points: multiple participants reported not understanding what earned points contributed to or unlocked, and this uncertainty is reflected in XP Rewarding being the lowest-rated Task 3 item (M = 5.17).
- No visual completion indicator: several participants relied solely on small "X of 9 lessons" text to judge progress; one participant re-clicked an already-completed lesson without realizing it, and another explicitly requested a visual (e.g., progress bar or checkmark) indicating completed lessons.
- Mismatched expectations for interactive elements: one participant expected the "investment simulator" control to launch a simulation, but it instead opened instructional content on how to buy stocks.
- Discoverability of the lessons tab: one participant rated ease of navigation lowest among all Task 3 respondents (a rating of 3), specifically citing difficulty finding the lessons tab itself, though this was not a widely shared experience.
- Small tap targets: one participant misclicked a small "learn lesson" button, which returned them to a previous screen.

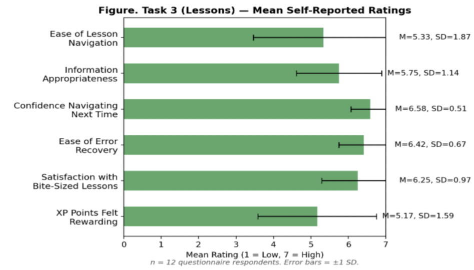

Figure 6: Task 3 (Lessons) mean self-reported ratings (±1 SD), N = 12. Scale: 1–7, higher = more favourable.

Table 1: Usability Testing Results Across Tasks

Metric	Task 1: Quiz	Task 2: Budget	Task 3: Lessons
Direct success	70% (7/10)	50% (5/10)	90% (9/10)
Indirect success	20% (2/10)	40% (4/10)	0% (0/10)
Failure / abandoned	0% (0/10)	0% (0/10)	0% (0/10)
Mean completion time	4.00 min (SD 2.24)	2.98 min (SD 2.23)	2.54 min (SD 1.55)
Mean satisfaction/ease rating	5.75 (SD 1.86)[1]	5.50 (SD 1.83)[2]	6.25 (SD 0.97)[5]
Full comprehension/Completed	90% (9/10)[3]	50% (5/10)[4]	100% (10/10)[6]

Observer-based rows use n = 10; questionnaire-based rows use N = 12.

[1]Quiz length appropriateness. [2]Overall ease of budget setup. [3]Understood quiz purpose (observer-rated). [4]Understood dashboard summary (observer-rated). [5]Satisfaction with bite-sized lessons. [6]Completed full lesson flow (observer-rated).

7.4 Summary of Results

All three tasks were completed, directly or indirectly, by every observed participant, with no failures or abandonment recorded in any task, indicating that the application's primary navigation was intuitive across the full user journey from onboarding through budgeting to financial education. However, the three tasks differed in the nature and severity of their usability challenges. The onboarding quiz primarily suffered from interaction-level issues related to answer selection, some of which are attributable to a known prototype constraint limiting each screen to a single interactive option. The budgeting task revealed more substantial usability problems involving dashboard interpretation, data consistency, and save-state feedback, and had the lowest Direct Success rate of the three tasks. The financial lessons task achieved the highest completion rate

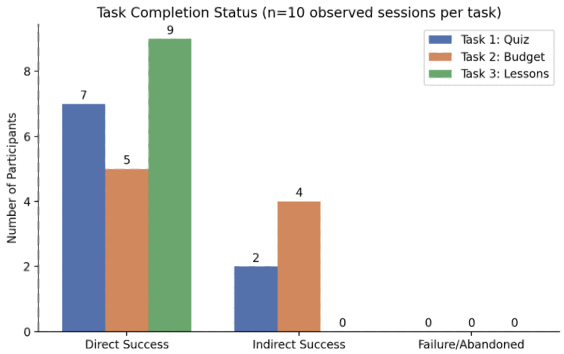

Figure 4. Task completion status by task, n = 10 observed sessions per task.

Figure 7: Task completion status by task, n = 10 observed sessions per task.

and generally the most favourable self-reported ratings, with its remaining friction concentrated in secondary feedback mechanisms, namely unclear XP/point purpose and the absence of a clear visual completion indicator, rather than in the core navigation or lesson content. These findings suggest that future design iterations should prioritize clarifying the budgeting dashboard, particularly category totals, spend values, and post-save confirmation, while making comparatively minor refinements to the quiz's interaction handling and the lessons section's progress and reward feedback.

8 Limitations

This study is subject to several limitations that should be considered when interpreting the findings. First, the evaluation was conducted on a Figma interactive prototype rather than a fully functional application. As a result, certain interactions, such as real-time budget updates, data persistence across sessions, or dynamic quiz-based recommendations, could only be simulated, which may have influenced participants' expectations and feedback in ways that would differ with a working product. Second, the sample size was small and drawn primarily from peers and classmates rather than a demographically diverse user base representative of Mino's target audience. This limits the generalizability of the findings, as participants' financial literacy levels, technical familiarity, and budgeting habits may not reflect the broader population the app is intended to serve. Third, testing was conducted in a single, controlled setting rather than in participants' natural environments, which may not fully capture how users would interact with a budgeting and financial literacy app in real-world, lower-attention contexts (e.g., quickly checking a budget while shopping). Fourth, the think-aloud method, while useful for surfacing real-time reactions, can alter natural user behaviour; participants may complete tasks more slowly or deliberately than they would unprompted, and self-reported reasoning may not always align with underlying cognitive processes. Finally, as this was a single round of usability testing, the findings reflect a snapshot of the prototype at one stage of development. Iterative changes based on these results have not yet been re-tested, so it is not yet known whether the proposed design recommendations fully resolve the issues observed.

9 Future Work

Based on the usability findings presented above, several improvements to the current prototype are warranted, prioritized below by where the data shows the clearest problems.

9.0.1 Budgeting Task (Highest Priority). The budgeting task had the lowest Direct Success rate of the three (50%, versus 70% for the quiz and 90% for lessons), and the findings point to specific, fixable causes rather than a vague sense of difficulty. Two participants independently reported that individual category totals did not match the overall spend figure shown on the dashboard, and "Spend Budget" columns displayed $0 for categories despite real amounts being entered, both of which undermine trust in the dashboard even though the underlying input mechanics work fine (80% of participants set category limits smoothly). More urgently, the save-state flow needs to be redesigned: after saving a budget, the prototype transitions to a blank "Edit Budget" screen with a "Clear Budget" button, and this ambiguity led two separate participants to accidentally clear their own budget data. Because this occurred independently for two participants, it should be treated as a reproducible design flaw rather than a one-off misclick, and prioritized accordingly.

- Fix the category-total calculation and $0 spend-value display bug on the dashboard.
- Redesign the post-save flow: add an explicit save confirmation and remove the destructive "Clear Budget" action from the screen shown immediately after saving.
- Label the budget period explicitly (weekly vs. monthly) at input time, since participants reported uncertainty about which timeframe their numbers applied to.
- Add colour-coding or other visual distinction between spending categories, as suggested by participants.
- Add a direct comparison between actual and expected spending per category to improve at-a-glance dashboard readability.

9.0.2 Onboarding Quiz. The quiz showed a different pattern: navigation and comprehension were not barriers (100% found the quiz entry point, 90% understood its purpose), and participants rated the controls themselves as easy to understand. The 50% of participants who struggled with quiz inputs were most likely running into the known prototype constraint described previously, where each screen exposed only one selectable option instead of the full set, rather than a genuine design problem; this should be verified once a fully interactive version is built. A few improvements are still worth making regardless of that constraint.

- Add the ability to revisit or edit a previous quiz answer, which participants expected and did not have.
- Reword the one question flagged as having two near-identical answer choices.
- Increase font size where flagged as a mild readability issue.
- Better communicate how the financial-personality archetype result is calculated — or let users see how it might shift as they use the app — given that perceived accuracy of the

archetype was the weakest-rated quiz dimension (M = 4.92 of 7).

9.0.3 Financial Lessons. The lessons performed best overall (90% Direct Success, highest satisfaction ratings), so remaining improvements here are refinements rather than fixes. “XP felt rewarding” was the lowest-rated item in this task (M = 5.17), with participants unclear on what earned points actually unlocked.

- Make the purpose of XP more visible and concrete.
- Replace the current small “X of 9 lessons” text with a visual completion indicator (a progress bar or checkmarks), since at least one participant re-clicked an already-completed lesson without realizing it.
- Rename or restructure the “investment simulator” entry point, since one participant expected it to launch an interactive simulation and it opened static instructional content instead, a naming mismatch rather than a comprehension failure.
- Enlarge tap targets on small buttons (e.g., “learn lesson”) to reduce accidental misclicks.
- Monitor for recurring discoverability issues with the lessons tab: one participant rated navigation as the most difficult among Task 3 respondents, though this was not a widely shared experience.

9.0.4 Broader Recommendations for Future Testing.

- Retest with a fully interactive prototype to isolate whether the quiz’s 50% input-struggle rate reflects the single-interactive-option constraint or a genuine design issue.
- Counterbalance task order in future studies: Task 2 and Task 3 were completed faster than Task 1, which may reflect participants growing more familiar with the prototype’s navigation over the course of a session rather than those tasks being inherently easier, and counterbalancing would help separate a real difficulty difference from a practice effect.
- Add more targeted debrief questions to future post-task questionnaires, since 0% of participants reported making an outright “mistake” on the budgeting task despite real, observed confusion around the dashboard, suggesting confusion that participants don’t attribute to their own error may otherwise go unsurfaced.
- Examine how the interface performs on smaller screen sizes, with a larger and more demographically varied participant pool, and with a longer testing window to see whether the archetype’s perceived accuracy improves as users watch it update over repeated use.

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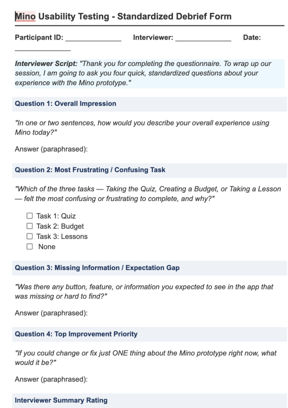
10 Appendices



Mino Usability Testing - Standardized Debrief Form

Participant ID: ______________ Interviewer: ______________ Date: ______________

Interviewer Script: *"Thank you for completing the questionnaire. To wrap up our session, I am going to ask you four quick, standardized questions about your experience with the Mino prototype."*

Question 1: Overall Impression

"In one or two sentences, how would you describe your overall experience using Mino today?"

Answer (paraphrased):

Question 2: Most Frustrating / Confusing Task

"Which of the three tasks — Taking the Quiz, Creating a Budget, or Taking a Lesson — felt the most confusing or frustrating to complete, and why?"

- ☐ Task 1: Quiz
- ☐ Task 2: Budget
- ☐ Task 3: Lessons
- ☐ None

Question 3: Missing Information / Expectation Gap

"Was there any button, feature, or information you expected to see in the app that was missing or hard to find?"

Answer (paraphrased):

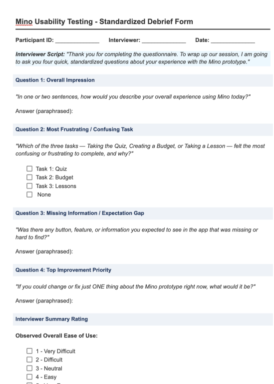
Question 4: Top Improvement Priority

"If you could change or fix just ONE thing about the Mino prototype right now, what would it be?"

Answer (paraphrased):

Interviewer Summary Rating

Figure 8: Usability Testing Debrief Form Page 1



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Answer (paraphrased):

Question 2: Most Frustrating / Confusing Task

"Which of the three tasks — Taking the Quiz, Creating a Budget, or Taking a Lesson — felt the most confusing or frustrating to complete, and why?"

- ☐ Task 1: Quiz
- ☐ Task 2: Budget
- ☐ Task 3: Lessons
- ☐ None

Question 3: Missing Information / Expectation Gap

"Was there any button, feature, or information you expected to see in the app that was missing or hard to find?"

Answer (paraphrased):

Question 4: Top Improvement Priority

"If you could change or fix just ONE thing about the Mino prototype right now, what would it be?"

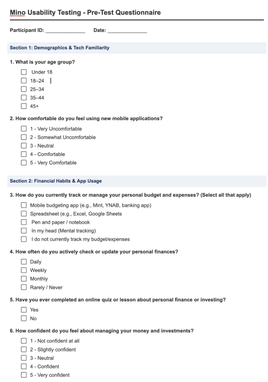
Answer (paraphrased):

Interviewer Summary Rating

Observed Overall Ease of Use:

- ☐ 1 - Very Difficult
- ☐ 2 - Difficult
- ☐ 3 - Neutral
- ☐ 4 - Easy
- ☐ 5 - Very Easy

Figure 9: Usability Testing Debrief Form Page 2



Mino Usability Testing - Pre-Test Questionnaire

Participant ID: ______________ **Date:** ______________

Section 1: Demographics & Tech Familiarity

1. What is your age group?

- ☐ Under 18
- ☐ 18–24
- ☐ 25–34
- ☐ 35–44
- ☐ 45+

2. How comfortable do you feel using new mobile applications?

- ☐ 1 - Very Uncomfortable
- ☐ 2 - Somewhat Uncomfortable
- ☐ 3 - Neutral
- ☐ 4 - Comfortable
- ☐ 5 - Very Comfortable

Section 2: Financial Habits & App Usage

3. How do you currently track or manage your personal budget and expenses? (Select all that apply)

- ☐ Mobile budgeting app (e.g., Mint, YNAB, banking app)
- ☐ Spreadsheet (e.g., Excel, Google Sheets
- ☐ Pen and paper / notebook
- ☐ In my head (Mental tracking)
- ☐ I do not currently track my budget/expenses

4. How often do you actively check or update your personal finances?

- ☐ Daily
- ☐ Weekly
- ☐ Monthly
- ☐ Rarely / Never

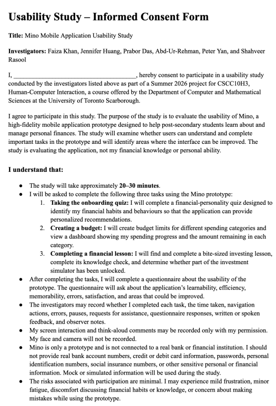
5. Have you ever completed an online quiz or lesson about personal finance or investing?

- ☐ Yes
- ☐ No

6. How confident do you feel about managing your money and investments?

- ☐ 1 - Not confident at all
- ☐ 2 - Slightly confident
- ☐ 3 - Neutral
- ☐ 4 - Confident
- ☐ 5 - Very confident

Figure 10: Usability Testing Pre Screening Questionnaire



Usability Study – Informed Consent Form

Title: Mino Mobile Application Usability Study

Investigators: Faiza Khan, Jennifer Huang, Prabor Das, Abd-Ur-Rehman, Peter Yan, and Shahveer Rasool

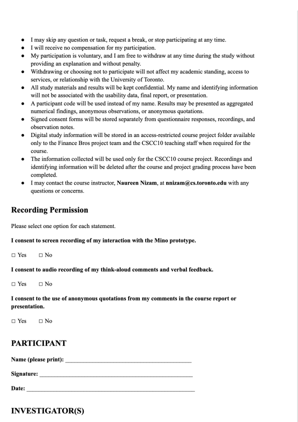
I, ______________________________________, hereby consent to participate in a usability study conducted by the investigators listed above as part of a Summer 2026 project for CSCC10H3, Human-Computer Interaction, a course offered by the Department of Computer and Mathematical Sciences at the University of Toronto Scarborough.

I agree to participate in this study. The purpose of the study is to evaluate the usability of Mino, a high-fidelity mobile application prototype designed to help post-secondary students learn about and manage personal finances. The study will examine whether users can understand and complete important tasks in the prototype and will identify areas where the interface can be improved. The study is evaluating the application, not my financial knowledge or personal ability.

I understand that:

- The study will take approximately **20–30 minutes**.
- I will be asked to complete the following three tasks using the Mino prototype:
  1. **Taking the onboarding quiz:** I will complete a financial-personality quiz designed to identify my financial habits and behaviours so that the application can provide personalized recommendations.
  2. **Creating a budget:** I will create budget limits for different spending categories and view a dashboard showing my spending progress and the amount remaining in each category.
  3. **Completing a financial lesson:** I will find and complete a bite-sized investing lesson, complete its knowledge check, and determine whether part of the investment simulator has been unlocked.
- After completing the tasks, I will complete a questionnaire about the usability of the prototype. The questionnaire will ask about the application's learnability, efficiency, memorability, errors, satisfaction, and areas that could be improved.
- The investigators may record whether I completed each task, the time taken, navigation actions, errors, pauses, requests for assistance, questionnaire responses, written or spoken feedback, and observer notes.
- My screen interaction and think-aloud comments may be recorded only with my permission. My face and camera will not be recorded.
- Mino is only a prototype and is not connected to a real bank or financial institution. I should not provide real bank account numbers, credit or debit card information, passwords, personal identification numbers, social insurance numbers, or other sensitive personal or financial information. Mock or simulated information will be used during the study.
- The risks associated with participation are minimal. I may experience mild frustration, minor fatigue, discomfort discussing financial habits or knowledge, or concern about making mistakes while using the prototype.

Figure 11: Usability Testing Informed Consent Form Page 1



- I may skip any question or task, request a break, or stop participating at any time.
- I will receive no compensation for my participation.
- My participation is voluntary, and I am free to withdraw at any time during the study without providing an explanation and without penalty.
- Withdrawing or choosing not to participate will not affect my academic standing, access to services, or relationship with the University of Toronto.
- All study materials and results will be kept confidential. My name and identifying information will not be associated with the usability data, final report, or presentation.
- A participant code will be used instead of my name. Results may be presented as aggregated numerical findings, anonymous observations, or anonymous quotations.
- Signed consent forms will be stored separately from questionnaire responses, recordings, and observation notes.
- Digital study information will be stored in an access-restricted course project folder available only to the Finance Bros project team and the CSCC10 teaching staff when required for the course.
- The information collected will be used only for the CSCC10 course project. Recordings and identifying information will be deleted after the course and project grading process have been completed.
- I may contact the course instructor, **Naureen Nizam**, at **nnizam@cs.toronto.edu** with any questions or concerns.

Recording Permission

Please select one option for each statement.

I consent to screen recording of my interaction with the Mino prototype.

☐ Yes ☐ No

I consent to audio recording of my think-aloud comments and verbal feedback.

☐ Yes ☐ No

I consent to the use of anonymous quotations from my comments in the course report or presentation.

☐ Yes ☐ No

PARTICIPANT

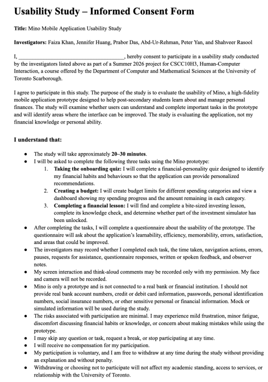
Name (please print): ______________________________

Signature: ______________________________

Date: ______________________________

INVESTIGATOR(S)

Figure 12: Usability Testing Informed Consent Form Page 2



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- The risks associated with participation are minimal. I may experience mild frustration, minor fatigue, discomfort discussing financial habits or knowledge, or concern about making mistakes while using the prototype.
- I may skip any question or task, request a break, or stop participating at any time.
- I will receive no compensation for my participation.
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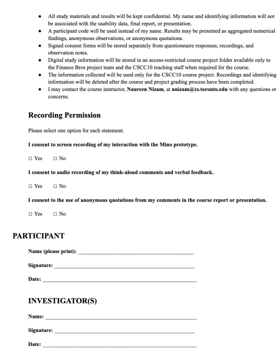
Figure 13: Usability Testing Informed Consent Form Page 3



- All study materials and results will be kept confidential. My name and identifying information will not be associated with the usability data, final report, or presentation.
- A participant code will be used instead of my name. Results may be presented as aggregated numerical findings, anonymous observations, or anonymous quotations.
- Signed consent forms will be stored separately from questionnaire responses, recordings, and observation notes.
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Recording Permission

Please select one option for each statement.

I consent to screen recording of my interaction with the Mino prototype.

☐ Yes ☐ No

I consent to audio recording of my think-aloud comments and verbal feedback.

☐ Yes ☐ No

I consent to the use of anonymous quotations from my comments in the course report or presentation.

☐ Yes ☐ No

PARTICIPANT

Name (please print): ______________________________

Signature: ______________________________

Date: ______________________________

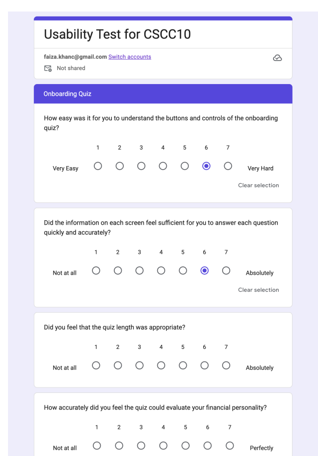
INVESTIGATOR(S)

Name: ______________________________

Signature: ______________________________

Date: ______________________________

Figure 14: Usability Testing Informed Consent Form Page 4



Usability Test for CSCC10

faiza.khanc@gmail.com Switch accounts

Not shared

Onboarding Quiz

How easy was it for you to understand the buttons and controls of the onboarding quiz?

Very Easy 1 2 3 4 5 6 7 Very Hard

Clear selection

Did the information on each screen feel sufficient for you to answer each question quickly and accurately?

Not at all 1 2 3 4 5 6 7 Absolutely

Clear selection

Did you feel that the quiz length was appropriate?

Not at all 1 2 3 4 5 6 7 Absolutely

How accurately did you feel the quiz could evaluate your financial personality?

Not at all 1 2 3 4 5 6 7 Perfectly

Figure 15: Usability Testing Post Test Questionnaire Page 1

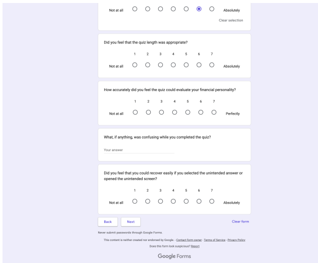


Figure 16: Usability Testing Post Test Questionnaire Page 2

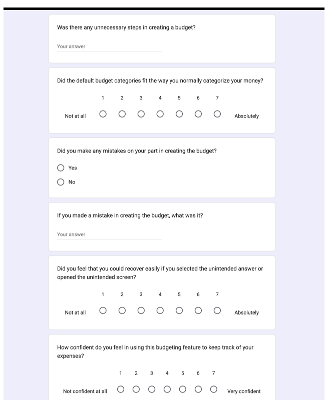


Figure 17: Usability Testing Post Test Questionnaire Page 3

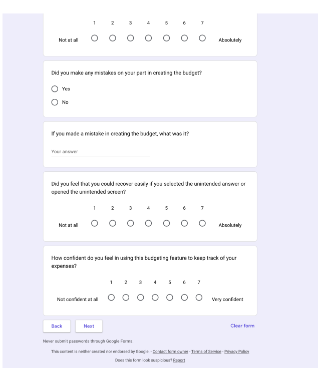


Figure 18: Usability Testing Post Test Questionnaire Page 4

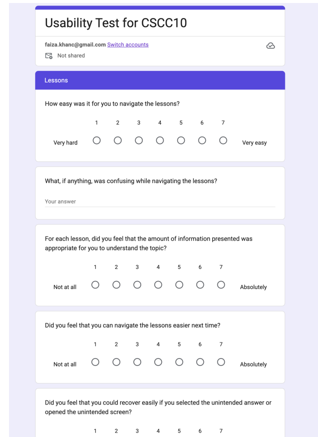


Figure 19: Usability Testing Post Test Questionnaire Page 5

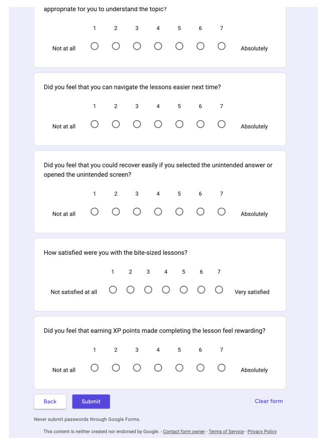


Figure 20: Usability Testing Post Test Questionnaire Page 6

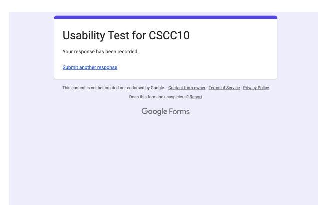


Figure 21: Usability Testing Post Test Questionnaire Page 7

Did you make any mistakes on your part in creating	If you made a mistake in creating the budget, what	Did you feel that you could recover easily if you sel
No	no	1
No		5
No	I don't think I made a mistake.	6
No	No mistakes	7
No		7
No		6
No	Nothing	6
No	N/A	7
No	N/A	4
No		4
No	I accidentally cleared the budget, but I didnt mean t	3
Yes	Yes, I may have accidentally cleared my entries.	4

Figure 27: Post Usability Testing Questionnaire, item 6

How confident do you feel in using this budgeting f	How easy was it for you to navigate the lessons?
2	1
6	7
5	6
7	7
1	7
7	4
6	5
7	6
6	5
5	3
4	7
6	6

Figure 28: Post Usability Testing Questionnaire, item 7

What, if anything, was confusing while navigating the lessons?
no
I wasn't sure what the points I earned in the end of the lesson went or contributed to.
Did not understand the concept of reward points
none
One thing that was a little confusing was then first going into the lesson, I started clicking the first lesson because I did not realize the first lesson was already completed, and I did not see anything that indicated this. I only realized that the top said that the first lesson was completed after clicking and failing on the first lesson.
The investment simulator in investments lesson confused me. I thought if I click investment simulator it would start a simulation through the lesson rather than teaching me how to buy stocks.
Something visually indicating that a lesson has been complete other than the 1 of 9 lessons text at the top of the page
It said I had completed 1 out of 9 but it wasn't clear which one I had already finished out of the 9. Also the "learn lesson'
finding the actual lesson tab itself. the location of it was hard to navigate.
Nothing
Nothing major.

Figure 29: Post Usability Testing Questionnaire, item 8

For each lesson, did you feel that the amount of inf	Did you feel that you can navigate the lessons easi	Did you feel that you could recover easily if you sel
6	6	7
7	7	6
6	6	5
6	6	7
7	7	7
3	7	6
6	7	7
5	7	7
7	7	7
5	6	6
5	7	6
6	6	6

Figure 30: Post Usability Testing Questionnaire, item 9

How satisfied were you with the bite-sized lessons'	Did you feel that earning XP points made completir	Q1: Age Group	Q2: Tech Comfort
7	6		
7	7	18–24	5 - Very Comfortable
6	4	18–24	4 - Comfortable
7	4	18–24	4 - Comfortable
5	2	18–24	5 - Very Comfortable
4	6	18-24	5 - Very Comfortable
7	7	18–24	5 - Very Comfortable
6	5	18–24	4 - Comfortable
7	4	18–24	3 - Neutral
7	4	18–24	3 - Neutral
6	7		
6	6	18–24	5 - Very Comfortable

Figure 31: Post Usability Testing Questionnaire, item 10

Q3: Budget Tracking Method	Q4: Finance Check Frequ	Q5: Prior Financial Lesso	Q6: Money Management	Task 1 Duration
In my head (Mental tracking)	Weekly	Yes	3 - Neutral	3 minutes
In my head (Mental tracking)	Monthly	No	4 - Confident	4 mins
Mobile budgeting app (e.g., Mint, YN	Weekly	No	4 - Confident	5 mins
I do not currently track my budget/e	Rarely / Never	No	5 - Very confident	2 minutes
In my head (Mental tracking)	Monthly	No	3 - Neutral	7 minutes
In my head (Mental tracking)	Daily	No	5 - Very confident	2 minutes 25 seconds
In my head (Mental tracking)	Weekly	No	3 - Neutral	2 minutes 23 seconds
In my head (Mental tracking)	Weekly	No	2 - Slightly confident	8 mins
Mobile budgeting app	Monthly	No	3 - Neutral	5 mins
In my head (Mental tracking)	Weekly	Yes	4 - Confident	1 minute 10 seconds

Figure 32: Post Usability Testing Questionnaire, item 11

Task 2 Duration	Task 3 Duration
7 minutes	1 minute
2 mins	2 mins
6 mins	5 mins
3 minutes	3 minutes
5 minutes	5 minutes
1 minute 33 seconds	1 minute 10 seconds
1 minute 33 seconds	1 minute 20 seconds
2 mins	3 mins
1 min	3 mins
41 seconds	52 seconds

Figure 33: Post Usability Testing Questionnaire, item 12

Debrief Q1: Overall Impression
Mino was very easy to navigate and used clear and concise language to display budgeting patterns.
it was a very smooth and easy experience. I think I understood most of it and didn't find it difficult to navigate at all. The only confusion I felt was in the budgeting screen where the category spend wasn't adding up to the total spend on top. Otherwise, no issues using the app itself.
Overall it felt clean and easy to pick up, once I understood the sidebar layout, moving between the quiz, budget, and lessons was quick and made sense.
I think the app is too cartoony which could deter potential users. The app also does not do anything inherently better than already existing banking apps.
Extremely straightforward and simple; the engaging UI used a soothing colour palette.
Really good. Polished. Would use if implemented.
Very good, clean design, good flow. Budgeting aspects are detailed, progress bars good. Small changes can be made visually in the 'create budget' page to make it more clear. For the budget history, having a colored bar visually breaking everything down; for the lessons, having a visual element showing what lessons were completed other than the 1 out of 9 text.
The UI was simple but informative. The comfy feel and cat characters had a very pleasant vibe but still felt professional in a corpor
Very cute! I think overall it was very visually pleasing and relatively easy to navigate.
Mino feels clean and easy to navigate overall; visual design is friendly, core flows all reachable without much hunting.

Figure 34: Post Usability Testing Questionnaire, item 13

Debrief Q2: Most Frustrating/Confusing Task	Debrief Q2 Explanation
None	N/A
None	N/A
Task 2: Budget	Wasn't sure if the budget amounts were meant to be weekly or monthly, and the "Miscella
Task 2: Budget	N/A
Task 3: Lessons	The 'Simulate Investment' button was slightly confusing because I initially thought it woul
Task 3: Lessons	Same as quesitonnaire answer
Task 2: Budget	Budgeting task. The most complex part is filling in the budget limits for each category.
Task 2: Budget	The budget was simple and easy to navigate but I feel that editing it can be a bit annoying
Task 2: Budget	The budget was simple and easy to navigate but I feel that editing it can be a bit annoying
Task 2: Budget	Entering income/expense figures took more back-and-forth; unclear which field to fill next

Figure 35: Post Usability Testing Questionnaire, item 14

Task 3: Completed Full Lesson Flow?	Task 3 Qualitative Notes	Observed Bottlenecks
Yes	Satisfying; Had struggles finding the workable categ	Some of the things were unclickable
Yes	N/A	Participant was struggling to underst
Yes	Hm, where do I find investment lessons specifically.	1. No back button in the quiz flow, pa
Yes	Read lesson outloud; Retook quiz; Considered other	Struggled getting through quiz flow
Yes	Covered stock ownership and price changes ($20 to	No main frustration observed.
Yes	None	Yes
Yes	There could be more visuals indicating progress.	None
Yes	Not recorded	Minor misclicks that were recoverabl
Yes	Not recorded	Not recorded
Yes	One knowledge-check question, Success screen, +2	Post-Save transition to blank "Edit Bu

Figure 41: Post Usability Testing Questionnaire, item 15

Observer Recommendations
Clear; Fix font size for alarming; Like the colours
Overall it was easy for them to follow the layout and find dedicated pages easily.
Recommend: (1) adding a back button or edit option within the quiz, (2) labeling
Appealing look; Looks fun; Similar to other banking apps; Cat mascot can restrict
Recommend clarifying the dashboard summary display; follow up directly with th
Good impression, suggested that the budgeting page could be more colorful.
More visuals.
Not recorded
Not recorded
Add save-confirmation state to budget flow; clarify/populate $0 spend-budget va

Figure 42: Post Usability Testing Questionnaire, item 16